GAJA SAVIYA - An Open-Data, Software-Only Early Warning and Safe-Routing Framework for Human-Elephant Conflict in Rural Sri Lanka

COMMON INTEGRATED SOLUTION REPORT

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Report submitted in partial fulfilment of the requirements for the

B.Sc (Hons) in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

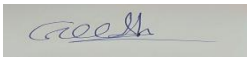
Sri Lanka

October 2026

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ABSTRACT

Human–elephant conflict (HEC) is a significant challenge in rural Sri Lanka, where interactions between humans and elephants can threaten human safety, damage property and affect livelihoods. This project proposes GAJA SAVIYA, an open-data, software-only early warning and safe-routing framework designed to support human safety in areas affected by human–elephant conflict. The prototype focuses on Anuradhapura, Sri Lanka, and integrates four complementary components: Seasonal Danger Map, Sparse Tracker, Safe Roads and Path Guard.

The Seasonal Danger Map identifies potential high-risk areas by combining relevant spatial and seasonal factors. Sparse Tracker estimates elephant movement between limited observation points and represents uncertainty in the estimated locations. Safe Roads assesses road-segment risk using available spatial information and proxy-based indicators. Path Guard uses route information to compare travel options and support the selection of routes that account for potential HEC-related risk as well as travel distance. Together, these components form an integrated framework intended to transform available data into risk information that can support situational awareness and safer route planning.

The proposed framework emphasises the use of publicly available data and software-based methods to reduce dependence on dedicated tracking hardware and specialised infrastructure. As a prototype, it also demonstrates how spatial risk assessment, movement estimation and risk-aware routing can be combined within a common system architecture. However, elephant movement estimates are simulated using published African elephant movement statistics rather than live tracking data from Sri Lankan elephants, and the Safe Roads component relies on proxy labels rather than a verified dataset of HEC incidents. Therefore, the framework should be interpreted as a research prototype rather than a validated operational early warning system.

GAJA SAVIYA provides a foundation for further research into accessible, data-driven approaches to human–elephant conflict mitigation. Future development should prioritise validation with reliable local elephant movement and HEC incident data, field-based evaluation, and assessment of the framework's effectiveness under real-world conditions.

Keywords: Human–elephant conflict, early warning system, open data, spatial risk assessment, elephant movement estimation, safe routing, Geographic Information Systems (GIS), Sri Lanka.

ACKNOWLEDGEMENTS

The project team would like to express its sincere gratitude to the project supervisor and the academic staff of the Department of Information Technology, Sri Lanka Institute of Information Technology (SLIIT), for their academic guidance and support throughout the development of this common integrated solution.

The team also acknowledges the organisations and communities that make publicly available data and technical resources accessible for research. These resources supported the investigation of human–elephant conflict and the design of the proposed software-based early warning and safe-routing framework.

The team appreciates the collaborative efforts of all members in developing the four components, establishing the integrated system architecture, and preparing this common report. The contributions of each member helped shape the overall solution and its intended approach to supporting human safety in areas affected by human–elephant conflict.

Finally, the team acknowledges the value of the academic and technical resources consulted during the project. The team accepts responsibility for the accuracy of the information presented, the appropriate acknowledgement of sources, and the limitations identified in this report.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
CRS	Coordinate Reference System
DEM	Digital Elevation Model
GIS	Geographic Information System
HEC	Human–Elephant Conflict
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
ML	Machine Learning
OSM	OpenStreetMap
SDG	Sustainable Development Goal
UI	User Interface
URL	Uniform Resource Locator
WGS 84	World Geodetic System 1984
XGBoost	Extreme Gradient Boosting

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Human–elephant conflict (HEC) is a significant challenge affecting both human communities and elephant conservation in Sri Lanka. Conflict can occur where elephant habitats overlap with agricultural land, villages and other areas of human activity. In rural areas, people may receive information about an elephant only after it has been sighted or damage has occurred. This creates a need for earlier, accessible and geographically relevant risk information.

Conventional approaches, including physical sensors and camera-based monitoring systems, may require equipment, maintenance and access to field locations. In addition, official elephant GPS collar records and detailed conflict incident databases may not be openly available to university research teams. These constraints create a practical challenge when developing and evaluating technology-based approaches to human–elephant conflict.

This project proposes an integrated, software-only early-warning and safe-routing framework for Anuradhapura District, Sri Lanka. It combines public geospatial datasets with controlled simulations to estimate geographic risk, represent elephant movement under infrequent location updates, identify potentially hazardous road sections and recommend alternative travel routes. The framework consists of four connected components: a seasonal danger map, a sparse movement tracker, a road-corridor risk and warning service, and Path Guard for safer route planning.

The danger map estimates risk by combining elephant habitat suitability, human exposure and a seasonal rainfall factor. The movement tracker uses an Extended Kalman Filter to estimate the location of a simulated elephant and represent uncertainty between updates. The road-risk component provides road-level warnings and explanations, while Path Guard compares the shortest route with an alternative that accounts for identified threats. These components are connected through a Next.js user interface.

The study is designed as an applied computational research project. It uses public data, project-generated simulations and software-based evaluation rather than field equipment or confidential Sri Lankan elephant tracking records. Consequently, its results must be interpreted as evidence about the prototype and its simulated scenarios, rather than proof of real-time tracking accuracy or operational effectiveness for wild elephants.

1.2 Stakeholders and user needs

The integrated framework is intended to address the information needs associated with human–elephant conflict. Potential stakeholders and their anticipated needs are presented in Table 1.1. These needs are derived from the project's stated purpose and should be validated through stakeholder engagement if such engagement is conducted.

Stakeholder group	Anticipated need
Residents of HEC-affected rural communities	Understand local risk areas and receive clear, location-specific warnings
Farmers and agricultural communities	Identify potentially hazardous areas near farmland and settlements
Road users and travellers	View potentially threatened road sections and compare travel routes
Wildlife and conservation authorities	Explore how open geospatial data could support supplementary risk assessment
Researchers and university teams	Develop and evaluate a repeatable approach without restricted tracking data or field equipment
Project evaluators and academic supervisors	Examine the methodology, integration, testing evidence and limitations of the prototype

Table 1.2- Stakeholders and user needs

The principal user need is to connect geographic risk information with understandable warnings and route alternatives. A static risk map alone does not provide the complete workflow envisioned by the project. The integrated solution therefore connects risk estimation, movement estimation, road-level advice and safer route planning.

Evidence limitation: The proposal does not document completed interviews, surveys or formal stakeholder validation. Therefore, the needs listed above are anticipated needs, not confirmed findings from users.

1.3 Alignment with the Sustainable Development Goals

The project's intended purpose may be associated with the United Nations Sustainable Development Goals (SDGs), particularly those concerned with sustainable communities, terrestrial ecosystems and responsible technological development.

- SDG 11 — Sustainable Cities and Communities: The framework explores how geographic risk information and route alternatives could contribute to safer decision-making in rural communities. This is an intended contribution; a reduction in accidents or injuries has not been established by the proposal.

- SDG 15 — Life on Land: The project focuses on human–elephant conflict and the relationship between elephant habitat and human activity. It investigates a technology-based approach that could support conservation-related decision-making without directly interacting with wild elephants.

The prototype's contribution to these goals is currently prospective. Further evidence, stakeholder assessment and real-world validation would be needed to establish whether the system delivers measurable social or conservation benefits.

1.4 Purpose and scope of the common report

This common integrated solution report presents the overall research problem, the proposed integrated framework, the relationships between its four components, the shared implementation approach and the evaluation of the complete system. Its purpose is to explain how the individual research and development activities contribute to a single software-based early-warning and safe-routing solution for human–elephant conflict in Anuradhapura District.

The report summarises the seasonal danger map, sparse elephant tracker, road-corridor risk and warning service, and Path Guard routing service. It also addresses component integration, shared interfaces, implementation, testing, sustainability, responsible innovation and future development.

Detailed component-specific methodologies and experiments should be presented in the corresponding individual theses, while this common report focuses on the complete system and the evidence supporting its integrated operation. The study is limited to public datasets, project-generated simulations and software-based testing. It does not claim to replace official wildlife monitoring systems or provide verified real-time locations of wild elephants.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

Human–elephant conflict (HEC) is a significant challenge in rural Sri Lanka, particularly in areas where elephant habitats overlap with agricultural land, settlements and human travel routes. People living and working in these areas may face risks when elephants move near villages, farms and roads.

One of the main challenges in reducing these risks is the limited availability of timely and accessible information about potential human–elephant conflict. Existing monitoring approaches may depend on physical sensors, GPS collars or official wildlife data that are difficult to access for a small-scale research project.

This project addresses the need for a software-based framework that integrates geographic risk assessment, elephant movement estimation, road-level risk warnings and route planning. The proposed system focuses on Anuradhapura District, Sri Lanka, and uses publicly available geographic and environmental data together with simulated elephant movement.

The overall problem is the lack of an integrated software framework that combines potential HEC risk estimation, movement tracking between infrequent location updates, road-risk assessment and risk-aware route planning within a single system.

The project proposes a four-component solution consisting of a Seasonal Danger Map, Sparse Tracker, Safe Roads and Path Guard. These components are designed to work together to demonstrate how publicly available data and computational models can support risk awareness and route decisions.

The system is an academic software prototype. Its outputs represent modelled risk and simulated movement rather than verified real-time elephant locations or guaranteed safe routes.

2.2 Root causes and overall research gap

Physical monitoring systems may require specialised equipment, installation, maintenance and field access. Official elephant-tracking records and detailed conflict-incident data may also be restricted. These limitations make it difficult to develop a prototype that relies entirely on continuous real-world monitoring.

The proposed system addresses this challenge by using publicly available datasets and simulated movement, enabling the development and testing of a software-based solution without requiring field equipment.

2.2.2 Limitations of Static Risk Maps

Geographic risk maps can identify areas where elephant habitat and human activity overlap. However, a static map does not necessarily represent seasonal changes or account for the changing estimated position of an elephant.

The proposed Seasonal Danger Map combines habitat suitability, human exposure and seasonal rainfall information to estimate potential risk across a grid covering Anuradhapura District.

2.2.3 Difficulty of Estimating Movement

When location updates are infrequent, an animal's position between updates must be estimated. Such estimates contain uncertainty, which can increase as the time between updates becomes longer.

The proposed Sparse Tracker uses simulated movement data and an Extended Kalman Filter (EKF) to estimate elephant positions between sparse location updates and represent positional uncertainty.

2.2.4 Lack of Road-Level Risk Information

Geographic risk maps alone do not provide detailed assessments of individual road segments. Road users need information that relates potential danger to the road network.

The proposed Safe Roads component combines geographic risk, road-network information and estimated elephant movement to assess road-segment risk and generate warnings.

2.2.5 Separation of Risk Assessment and Route Planning

A route that minimises distance may pass through areas with higher estimated risk. Therefore, route planning should consider potential danger alongside travel distance.

The proposed Path Guard component compares a shortest route with a risk-aware alternative using road-network and risk information.

2.2.6 Research Gap

The identified research gap is the need for a unified, software-only framework that connects seasonal geographic risk estimation, simulated elephant movement tracking, road-level risk assessment and risk-aware route planning using public data.

The project addresses this gap by integrating the four components into a single prototype and evaluating their technical behaviour through model testing and simulation.

2.3 Subproblem and component mapping

The overall problem is divided into four subproblems, each addressed by a specific component.

2.3.1 Component 1 — Seasonal Danger Map

This component addresses the problem of identifying areas with potentially higher human–elephant conflict risk. It uses public elephant occurrence records, land-cover information, population data and seasonal climate information.

The proposed approach uses an XGBoost model to estimate habitat suitability and combines this with human exposure and a seasonal rainfall multiplier to produce grid-based risk scores.

The component provides geographic risk information for use by the road-risk assessment component.

2.3.2 Component 2 — Sparse Tracker

This component addresses the problem of estimating elephant movement when location updates are infrequent. It uses a simulated elephant route and published movement statistics to demonstrate an Extended Kalman Filter.

The component estimates positions between updates and represents positional uncertainty. Its outputs are used by the road-risk component to assess the proximity of road segments to the estimated movement area.

2.3.3 Component 3 — Safe Roads

This component addresses the problem of assessing risk at road-segment level. It combines geographic risk information, estimated elephant position and uncertainty, and road-network data.

The proposed XGBoost model estimates proxy road risk and supports the generation of road-level warnings. The component provides road-risk information for route planning.

2.3.4 Component 4 — Path Guard

This component addresses the problem of incorporating modelled risk into route planning. It uses road-network and risk information to compare a shortest route with a risk-aware alternative.

The component is intended to connect the preceding components to a unified user interface, allowing the system's available risk and routing information to be presented together.

Component	**Subproblem**	**Main output**
Component 1: Seasonal Danger Map	Seasonal geographic risk estimation	Grid-based risk scores and spatial layers
Component 2: Sparse Tracker	Movement estimation between sparse updates	Estimated position and uncertainty
Component 3: Safe Roads	Road-segment risk assessment	Road-risk estimates and warnings
Component 4: Path Guard	Risk-aware route planning	Shortest and alternative routes

Table 2.3- Mapping of subproblems to system components

The four components form an integrated workflow in which geographic risk and movement estimates support road-risk assessment, and road-risk information supports route planning.

2.4 Overall aim and group objectives

2.4.1 Overall Aim

The overall aim of the project is to design, develop and test a software-only prototype for Anuradhapura District that uses publicly available data and simulated elephant movement to estimate potential human–elephant conflict risk, assess road-level risk and demonstrate risk-aware route planning.

2.4.2 Group Objectives

The project has the following objectives:

1. To develop a seasonal geographic risk map using publicly available elephant occurrence, land-cover, population and climate data.
2. To implement and evaluate an Extended Kalman Filter for estimating simulated elephant positions between infrequent location updates.
3. To develop a road-segment risk assessment model that combines geographic risk, road-network information and estimated elephant movement.
4. To implement route planning that compares the shortest route with a risk-aware alternative.
5. To integrate the four components into a unified software prototype.
6. To evaluate the technical performance, integration and limitations of the overall system using documented tests and simulation results.

Objective	Evaluation evidence
Seasonal risk mapping	Model evaluation and inspection of geographic risk outputs
Movement estimation	Tracking-error measurements against the simulated reference path
Road-risk assessment	Road-segment outputs, warnings and model evaluation
Risk-aware routing	Route comparisons, distance differences and risk-exposure measurements
System integration	End-to-end tests and demonstration of component communication

Overall evaluation	Consolidated test results and documented limitations

Table 2.4.2- Objectives and evaluation evidence

2.4.3 Research Questions

The project addresses the following research questions:

- RQ1: Can public elephant sightings, land-use data, population data and climate information produce a geographically meaningful HEC proxy-risk map for Anuradhapura District that changes with the season?
- RQ2: How accurately can a constant-velocity EKF estimate the hidden route of a simulated elephant when it receives only hourly location updates?
- RQ3: Can a road-corridor risk model and distance from the tracker's uncertainty area produce useful road warnings, and can risk-aware routing reduce travel through highly threatened road sections?
- RQ4: Can the four components operate as an integrated system that users can understand during a demonstration using recorded simulation playback?

2.5 Scope assumptions and constraints

2.5.1 Project Scope

The project focuses on developing and testing a software-only prototype for Anuradhapura District, Sri Lanka.

The scope includes:

- Collection and preparation of publicly available geographic and environmental datasets.
- Development of a 500-metre grid-based seasonal risk map.
- Habitat-suitability modelling using XGBoost.
- Combination of habitat suitability, human exposure and seasonal rainfall information.

- Simulation of elephant movement using published movement statistics.
- Position estimation using an Extended Kalman Filter.
- Representation of movement uncertainty.
- Road-network preparation and road-segment risk assessment.
- Generation of road-level risk warnings.
- Comparison of shortest and risk-aware alternative routes.
- Integration of the four components through a unified interface.
- Technical testing and evaluation of the prototype.

2.5.2 Project Exclusions

The project does not include:

1. Capturing, handling or collaring wild elephants.
2. Installing physical sensors or field-monitoring equipment.
3. Claiming access to restricted official elephant-tracking records.
4. Claiming that simulated movement demonstrates real-time tracking accuracy for actual elephants.
5. Treating proxy risk scores as verified probabilities of HEC incidents.
6. Guaranteeing that a recommended route is safe.
7. Replacing official wildlife monitoring or warning services.

2.5.3 Assumptions

The project is based on the following assumptions:

- Public datasets provide sufficient information to demonstrate the proposed framework.
- Geographic datasets can be aligned to a compatible coordinate reference system.

- Published elephant movement statistics can support a controlled movement simulation.
- A known simulated route can be used to evaluate tracking estimates.
- Proxy labels can support the development and demonstration of the road-risk model.
- The four components can exchange compatible data through defined interfaces.
- The available road network is suitable for demonstrating route calculation within the study area.

These assumptions affect the interpretation of the results and must be considered during evaluation.

2.5.4 Technical Constraints

The prototype is subject to several limitations. Public elephant occurrence records may have incomplete geographic coverage. Published movement statistics from African elephants may not accurately represent the movement of Asian elephants in Sri Lanka. The road-risk model uses proxy labels rather than a comprehensive dataset of confirmed HEC incidents.

In addition, movement estimates become less certain as the interval between updates increases. Errors in geographic processing, data exchange or road-network preparation may affect the warnings and routes generated by the integrated system.

Consequently, successful software testing can demonstrate that the prototype performs its specified computational functions, but it cannot independently establish that the system reduces actual human–elephant conflict incidents.

2.5.5 Ethical Considerations

The project uses public datasets and simulated movement to avoid direct intervention with wild elephants. The system must clearly distinguish between observed public records, simulated movement and model-generated estimates.

Risk warnings and route recommendations should be presented as decision-support information rather than guarantees of safety. The team must also consider the permitted use of datasets and mapping resources and handle any user location information collected during testing responsibly.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

The proposed solution is an integrated, software-only early-warning and safe-routing framework designed to address Human–Elephant Conflict (HEC) in rural Sri Lanka. The system focuses on the Anuradhapura District and combines publicly available environmental, geographical, population, and wildlife data with simulated elephant movement to estimate potential risk areas and support safer route selection.

The integrated system consists of four main components: the Seasonal Danger Map, Sparse Tracker, Safe Roads, and Path Guard. Each component addresses a specific part of the overall problem and exchanges relevant information with the other components to support the complete workflow.

The Seasonal Danger Map estimates potential HEC risk using habitat suitability, human exposure, and seasonal rainfall information. The Sparse Tracker estimates elephant positions between infrequent location updates and represents positional uncertainty. Safe Roads combines spatial risk, movement estimates, and road-network information to assess potential road-segment risk. Path Guard uses the resulting risk information to compare route options and provide risk-aware route guidance.

The overall solution is intended to demonstrate how publicly available data, machine-learning models, movement estimation, and route-planning algorithms can be combined within a single software application. As a prototype, the system does not replace official wildlife monitoring services and cannot guarantee that a recommended route is completely safe.

3.2 High level architecture

The high-level architecture is organised into five main layers: the data layer, integrated processing layer, integration layer, application layer, and end-user layer.

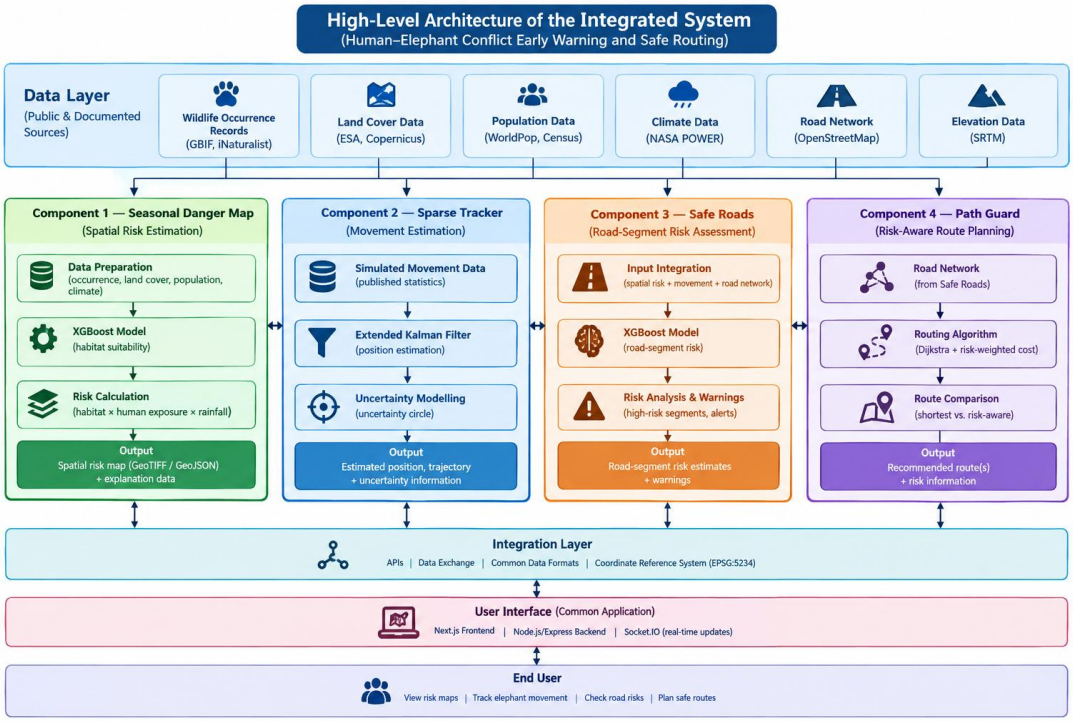


Figure 3.1- Overall System Architecture of GAJA SAVIYA

3.2.1 Data Layer

The data layer provides the information required by the four components. The proposed sources include GBIF and iNaturalist wildlife occurrence records, ESA WorldCover land-cover data, WorldPop population data, NASA POWER climate information, OpenStreetMap road networks, and elevation data from Copernicus GLO-30 or other documented elevation sources. Published elephant GPS movement statistics are used to inform simulated movement.

These datasets are processed into suitable formats before being used for risk modelling, movement estimation, road analysis, and route planning.

3.2.2 Integrated Processing Layer

The integrated processing layer contains the four main components.

- Component 1 — Seasonal Danger Map: Estimates spatial HEC risk using habitat suitability, human exposure, and seasonal rainfall.

- Component 2 — Sparse Tracker: Estimates elephant movement and positional uncertainty from simulated routes and sparse location updates.

- Component 3 — Safe Roads: Estimates road-segment risk using spatial risk, movement information, and road-network characteristics.

- Component 4 — Path Guard: Compares shortest-distance and risk-aware routes using the road network and risk information.

Each component produces outputs that contribute to the overall system.

3.2.3 Integration Layer

The integration layer enables the components to exchange information through defined interfaces and compatible data formats. Spatial risk maps, estimated positions, uncertainty information, road-segment risk estimates, and route data must be represented consistently to support reliable communication.

The proposed architecture includes a FastAPI service for spatial risk outputs and a Node.js/Express backend for application-level operations. The road-network data used by Path Guard is intended to remain consistent with the data processed by Safe Roads.

3.2.4 Application Layer

The proposed common user interface uses Next.js to present information from the integrated components. Node.js/Express and Socket.IO support backend communication and updates where implemented.

The interface is intended to allow users to view potential risk areas, inspect estimated elephant movement, identify potentially hazardous road sections, and compare route options.

3.2.5 End-User Layer

The end-user layer represents people who may use the prototype to explore risk information and route alternatives. Users can select a starting point and destination within the supported study area and view route options alongside relevant risk information.

The system provides decision-support information rather than a guarantee of safety. Its usefulness depends on the quality of available data, the modelling assumptions, and the performance of the implemented prototype.

3.3 Component 1 summary

The Seasonal Danger Map estimates the spatial distribution of potential human–elephant conflict risk within Anuradhapura District. It combines wildlife occurrence records with environmental, land-cover, population, and seasonal climate information to identify areas that may require greater attention.

The proposed approach divides the study area into a 500-metre grid containing 28,817 cells, using the coordinate reference system EPSG:5234. GBIF and iNaturalist records provide elephant occurrence information for habitat-suitability modelling. Land-cover and population datasets represent human exposure, while NASA POWER data provides seasonal rainfall information.

An XGBoost model is proposed to estimate habitat suitability. The resulting suitability estimate is combined with human exposure and a seasonal rainfall multiplier to calculate a potential risk score:

$$\textbf{Risk Score} = \textbf{Habitat Suitability} \times \textbf{Human Exposure} \times \textbf{Seasonal Rainfall Multiplier}$$

The component is designed to generate spatial outputs, including GeoTIFF and GeoJSON risk maps, together with explanation data that helps users understand the

factors influencing estimated risk. SHAP-based explanations are proposed to support model interpretability.

The main output is a spatial risk layer that can be consumed by Safe Roads and the integrated application. The risk map represents modelled potential risk rather than confirmed or real-time elephant presence.

3.4 Component 2 summary

The Sparse Tracker estimates elephant movement when continuous location data is unavailable. The prototype uses simulated elephant routes informed by published African elephant GPS movement statistics rather than assuming access to live tracking data from elephants in Sri Lanka.

An Extended Kalman Filter (EKF) is proposed to estimate the elephant's position between sparse observations. The filter predicts movement using the current state estimate and updates the estimate when a new observation becomes available.

In addition to the estimated position, the component calculates positional uncertainty and represents it through an uncertainty circle. This information allows downstream components to consider not only the estimated location but also the uncertainty associated with that estimate.

The principal outputs are the estimated position, movement trajectory, and positional uncertainty. These outputs are supplied to Safe Roads to support the assessment of road sections near potentially hazardous movement areas.

Evaluation is based on simulated routes with known positions. Therefore, any reported accuracy measures apply to the simulated evaluation and should not be interpreted as evidence of equivalent accuracy in real-world elephant tracking.

3.5 Component 3 summary

Safe Roads estimates potential HEC risk along road segments by combining the spatial risk map, movement estimates, and road-network information.

The component uses road-network data from OpenStreetMap and processes it into road segments suitable for analysis. Proposed model inputs include road type, local environmental risk, time-related variables, and the distance or overlap between road segments and the Sparse Tracker's uncertainty circle.

An XGBoost model is proposed to estimate road-segment risk. The training labels are based on proxy indicators, such as high-risk corridors and areas near farmland or settlements. These proxies do not represent verified human–elephant incident records and must be interpreted accordingly.

The component is designed to classify road sections according to their estimated risk and provide understandable warnings. Road sections near an estimated movement area or in locations with exceptionally high environmental risk may be highlighted for caution.

Safe Roads supplies road-segment risk estimates and associated warning information to Path Guard. This enables route planning to consider potential risk in addition to travel distance.

A risk-weighted route cost may be calculated using:

$$\boldsymbol{C = L(1 + \lambda R)}$$

where $\boldsymbol{C}$ is the route cost, $\boldsymbol{L}$ is the road-segment length, $\boldsymbol{R}$ is the estimated segment risk, and $\boldsymbol{\lambda}$ controls the influence of risk on route selection.

3.6 Component 4 summary

Path Guard is the routing and application-level component of the integrated system. It uses the road network and risk information produced by the other components to compare route options and support risk-aware navigation.

The proposed implementation uses Node.js/Express for backend operations and Socket.IO for communication and updates where required. The component uses a local JSON representation of the road network that is intended to remain consistent

with the network used by Safe Roads. A graph-based routing algorithm, such as Dijkstra's algorithm, is proposed for calculating route options.

Path Guard compares a shortest-distance route with an alternative that accounts for estimated road-segment risk. Risk-weighted costs and movement-related threat areas can be considered when evaluating candidate routes.

The component is intended to provide users with recommended routes and relevant risk information through the common application interface. Its outputs help users compare route alternatives, but the system cannot guarantee that a route will avoid all encounters because the underlying risk and movement estimates are subject to uncertainty.

Path Guard completes the integrated workflow by converting the outputs of the risk-mapping, movement-estimation, and road-risk components into route-planning information.

3.7 End to end user and data flow

The end-to-end workflow explains how information moves from the source datasets through the four components to the final user interface.

3.7.1 Data Processing Flow

1. Data collection: Public wildlife occurrence records, land-cover information, population data, climate data, elevation information, and road-network data are collected from the proposed sources.
2. Spatial risk estimation: The Seasonal Danger Map processes the relevant datasets and produces a spatial risk map with associated explanatory information.
3. Movement estimation: The Sparse Tracker processes simulated movement and sparse observations to estimate position, trajectory, and positional uncertainty.

4. Road-risk assessment: Safe Roads combines the spatial risk map, movement estimates, and road-network features to estimate risk along road segments.
5. Route calculation: Path Guard uses the road network and road-risk information to calculate and compare shortest-distance and risk-aware route options.
6. Presentation: The common user interface presents available risk maps, movement estimates, road-risk information, and route alternatives to the user.

3.7.2 User Interaction Flow

The intended user interaction begins when a user opens the application and views the available spatial risk information. The user can then inspect the estimated movement information and the risk indicators associated with relevant road sections. When route planning is required, the user selects an origin and destination, after which Path Guard calculates route alternatives using the available road-network and risk data.

The application presents the resulting routes and their associated risk information to support comparison and decision-making.

3.7.3 Component Data Exchange

Source component	**Information exchanged**	**Receiving component**
Seasonal Danger Map	Spatial risk map and explanation data	Safe Roads
Sparse Tracker	Estimated position, trajectory, and uncertainty	Safe Roads

Safe Roads	Road network and road-segment risk information	Path Guard
Path Guard	Route alternatives and associated risk information	Common user interface

Table 3.7.3- Component Data Exchange

The integrated workflow depends on compatible data formats, consistent spatial references, and clear interface definitions. If a component's output is unavailable or invalid, the application should avoid presenting the resulting risk estimate or route as fully reliable.

Overall, the end-to-end design connects environmental risk estimation, movement estimation, road-risk assessment, and route planning into one software-based framework. This structure supports the project's central objective of demonstrating an integrated approach to HEC risk awareness and safer route selection using public data and simulated movement.

The intended data flow is summarised below.

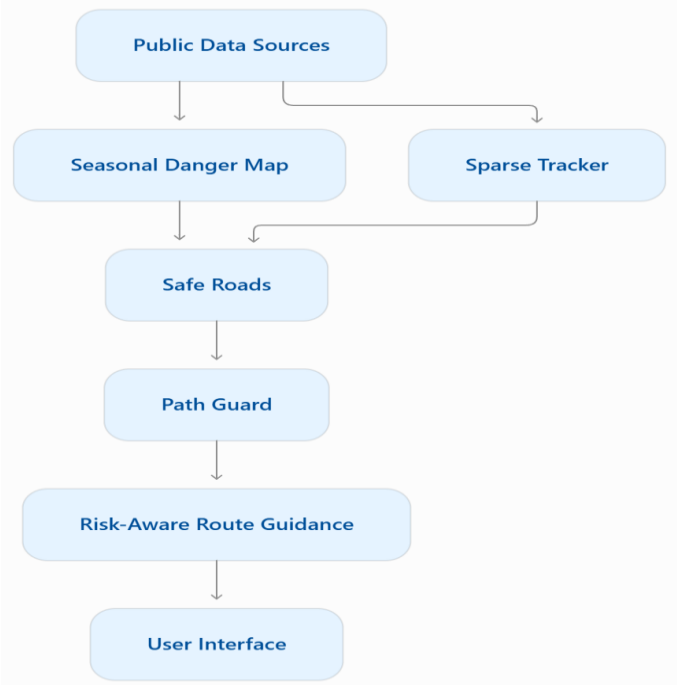


Figure 3.2- End-to-End Information Flow

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

The proposed Human–Elephant Conflict Early Warning and Safe Routing System integrates four components into a single software-based framework: Seasonal Danger Map, Sparse Tracker, Safe Roads, and Path Guard. The integration strategy focuses on ensuring that the outputs of each component can be used by the next component to support the overall objective of identifying potential HEC risk areas and providing risk-aware route guidance.

A modular integration approach is proposed so that each component can perform its specific processing tasks while exchanging information through defined interfaces. This approach supports independent component development, simplifies testing, and allows individual components to be updated without redesigning the entire system.

The integration process follows the sequence below.

1. Spatial risk integration: The Seasonal Danger Map generates a spatial risk map using wildlife occurrence, land-cover, population, and seasonal climate data.

2. Movement information integration: The Sparse Tracker generates estimated elephant positions, movement trajectories, and positional uncertainty from simulated movement data.

3. Road-risk integration: Safe Roads combines the spatial risk map, movement information, and road-network data to estimate potential risk along road segments.

4. Routing integration: Path Guard uses the road network and estimated road-segment risk to compare the shortest-distance route with risk-aware alternatives.

5. Application integration: The common user interface presents risk maps, movement estimates, road-risk information, and route recommendations.

The proposed architecture uses FastAPI to provide access to spatial risk outputs and Node.js/Express for application-level backend operations. Next.js is proposed for the common frontend, while Socket.IO supports communication and updates where required.

Integration must also maintain consistency between geographical datasets, coordinate reference systems, road-network representations, and risk values. These requirements are particularly important because an inconsistency in one component may affect the outputs of downstream components.

The integration strategy is based on the proposed architecture. The exact implementation status of individual interfaces and services remains subject to verification against the completed prototype.

4.2 Interfaces data contracts and shared models

Interfaces and data contracts define how information is transferred between components. Each component must provide its outputs in a consistent structure so that receiving components can interpret the data correctly.

The integration design uses spatial data formats such as GeoTIFF and GeoJSON, structured JSON data for movement and road information, and application programming interfaces (APIs) for communication between software services.

4.2.1 Component Interfaces

Producing component	**Data exchanged**	**Receiving component**	**Purpose**
Seasonal Danger Map	Spatial risk map and explanation data	Safe Roads	Supplies geographical risk information for road-risk assessment
Sparse Tracker	Estimated position, trajectory, and uncertainty	Safe Roads	Identifies road sections near estimated movement areas
Safe Roads	Road network and road-segment risk estimates	Path Guard	Supplies road and risk information for route calculation

Path Guard	Route alternatives and risk information	Common user interface	Presents route guidance to users

Table 4.2.1- Component Interfaces

4.2.2 Shared Data Models

The integrated system requires common representations for spatial risk, movement estimates, road segments, and route alternatives.

Spatial risk model: The spatial risk model represents a geographical area and its estimated HEC risk. It may include a grid-cell identifier, risk score, spatial geometry or raster-cell reference, and relevant explanatory information. GeoTIFF is intended for raster-based spatial outputs, while GeoJSON supports geographical features and associated attributes.

Movement estimate model: The movement model represents an estimated elephant position and its associated uncertainty. It may contain a timestamp, coordinates, trajectory information, and an uncertainty measure. These fields allow Safe Roads to compare estimated movement areas with road locations.

Road-segment model: The road-segment model represents a section of the road network and its associated risk information. Relevant attributes may include a segment identifier, road geometry, road type, segment length, estimated risk score, and warning category.

Route model: The route model represents a calculated route between an origin and destination. It may contain an ordered sequence of road segments, total route distance, estimated route cost, and associated risk information.

These models describe the information required for integration. Their exact field names, validation rules, and serialisation formats must be confirmed against the implemented code before they are treated as final interface specifications.

4.2.3 Coordinate Reference System and Data Consistency

Spatial consistency is essential because the system combines risk-grid cells, movement coordinates, and road geometries. The Seasonal Danger Map proposes a 500-metre grid using EPSG:5234. Other spatial datasets must be transformed into compatible coordinate systems when necessary before spatial calculations are performed.

The system should ensure that:

- Coordinates are interpreted using the correct coordinate reference system.
- Risk scores use consistent definitions and expected value ranges.
- Road-segment identifiers remain consistent between Safe Roads and Path Guard.
- Missing, invalid, or outdated data is identified before processing.
- Movement timestamps and associated uncertainty information are represented consistently.

These checks reduce the possibility of incorrect risk calculations caused by incompatible data representations.

4.2.4 API and Communication Interfaces

FastAPI is proposed for serving spatial risk outputs, while Node.js/Express supports application-level operations and communication with the routing functionality. Socket.IO may be used to transmit updates to the interface when required by the implementation.

The interfaces should provide clear input and output definitions and return appropriate error information when requests cannot be processed. API endpoints, request formats, response formats, and authentication requirements must be documented according to the actual implementation.

4.3 Shared nonfunctional requirements

The integrated system requires shared non-functional requirements to ensure that the four components operate consistently and provide understandable outputs.

Requirement	Description	Application to the integrated system
Performance	The system should process inputs and return outputs within a reasonable time for prototype demonstrations.	Risk-map retrieval, road-risk processing, and route calculation should be evaluated for response time.
Reliability	Components should handle invalid inputs and unavailable services without producing misleading results.	Errors in risk or movement data should be identified before route recommendations are generated.
Data consistency	Shared geographical and risk data should use compatible formats and definitions.	Spatial risk, movement coordinates, and road-network information must align correctly.
Interoperability	Components should exchange information through documented interfaces.	APIs and structured data formats should support communication between the components.
Usability	Users should be able to interpret risk information and compare route alternatives.	Maps, warnings, and route information should be presented clearly.
Maintainability	Individual components should be capable of being modified and tested independently.	Modular services and documented interfaces should reduce unnecessary dependencies.
Scalability	The architecture should allow future expansion to additional geographical areas or datasets.	Data-processing and routing components should be designed to accommodate future extensions.

Security	Application interfaces and data-processing services should be protected against unauthorised access where applicable.	Deployment-specific access controls and secure configuration should be considered.
Explainability	Risk information should be understandable to users.	Spatial risk explanations and road-risk indicators should help communicate why an area or route is flagged.
Responsible use	Outputs should not be represented as guaranteed predictions or verified real-time warnings.	The interface and report should communicate uncertainty and the prototype's limitations.

Table 4.3- Shared nonfunctional requirements

These requirements define the qualities expected of the integrated system. Quantitative performance targets and acceptance thresholds should be specified during testing based on the prototype's capabilities and evaluation needs.

4.4 Integrated workflow and exception handling

The integrated workflow connects the processing stages so that the final route recommendation uses the available spatial, movement, and road-risk information.

4.4.1 Integrated Workflow

1. Initialisation: The application loads the required configuration and makes the available data services accessible.
2. Risk-map processing: The Seasonal Danger Map provides the spatial risk layer and associated explanatory information.
3. Movement processing: The Sparse Tracker estimates movement and positional uncertainty from the available simulated observations.
4. Road-risk processing: Safe Roads combines spatial risk, movement estimates, and road-network attributes to calculate road-segment risk.

5. Route planning: Path Guard calculates the shortest-distance route and a risk-aware alternative using the available road-network and risk information.
6. Result presentation: The common interface presents the available route alternatives and relevant risk information.
7. Error checking: If a required input or service is unavailable, the application should communicate the limitation instead of presenting incomplete results as fully reliable.

4.4.2 Exception Handling

Exception handling is important because the integrated system depends on multiple data sources and software services. Errors in one component may affect other components and the final route recommendation.

Exception	**Potential effect**	**Intended handling**
Risk-map data unavailable	Safe Roads cannot access the expected spatial risk layer.	Report the unavailable data and avoid claiming that risk-based assessment is complete.
Missing movement estimate	Road-risk calculations cannot use the expected movement information.	Identify the missing input and indicate that movement-based warnings are unavailable.
Invalid spatial coordinates	Spatial comparisons may produce incorrect results.	Validate coordinate values and coordinate reference systems before processing.
Road-network data unavailable	Path Guard cannot calculate a valid route.	Display an appropriate error and request valid road-network data.
Invalid or missing risk values	Route costs may be calculated incorrectly.	Validate risk values and reject or flag invalid records.

No route found	The requested origin and destination cannot be connected using the available network.	Inform the user that no route could be calculated.
API or service failure	Components may be unable to exchange information.	Return an appropriate error and prevent incomplete outputs from being presented as successful results.
Delayed or uncertain movement data	Risk estimates may not represent current conditions.	Display relevant data limitations and avoid implying that the movement estimate is a verified live location.

Table 4.4.2-Exception cases and expected handling

Exception handling should prioritise clear communication and prevent unsupported safety claims. The specific recovery mechanisms, logging arrangements, and fallback behaviour must be confirmed during implementation and testing.

4.5 Team contribution and dependency matrix

The project is divided into four technical components that contribute to the integrated solution. Each component has a defined role and depends on outputs from other parts of the system.

Component	**Main responsibility**	**Main output**	**Integration dependency**
Component 1 — Seasonal Danger Map	Spatial HEC risk estimation	Spatial risk map and explanation data	Supplies geographical risk information to Safe Roads

Component 2 — Sparse Tracker	Movement estimation and uncertainty modelling	Estimated position, trajectory, and uncertainty	Supplies movement information to Safe Roads
Component 3 — Safe Roads	Road-segment risk assessment	Road-risk estimates and road-network information	Uses Components 1 and 2; supplies information to Path Guard
Component 4 — Path Guard	Risk-aware route planning and application-level integration	Route alternatives and risk information	Uses road-network and risk outputs from Safe Roads

Table 4.5- Team contribution and dependency matrix

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The proposed Human–Elephant Conflict Early Warning and Safe Routing System uses a modular software architecture that combines machine learning, geospatial data processing, movement estimation, road-network analysis, and web-based route planning. The technology stack is designed to support the integration of the four components into a common application.

The proposed technologies and their intended purposes are summarised below.

Technology or tool	Purpose in the integrated system
Python	Data processing, machine-learning development, and analytical operations
XGBoost	Habitat-suitability modelling and road-segment risk estimation
Extended Kalman Filter (EKF)	Estimating elephant positions between sparse observations
GeoTIFF	Storing and exchanging raster-based spatial risk information
GeoJSON	Representing geographical features and spatial outputs
OpenStreetMap	Providing road-network data for road-risk assessment and routing
FastAPI	Serving spatial risk outputs and supporting communication with analytical services
Node.js and Express	Supporting backend application operations and routing integration
Next.js	Providing the common web-based user interface
Socket.IO	Supporting application updates and communication where required
Streamlit	Supporting experimental development and visualisation of the Seasonal Danger Map

Table 5.1- Common technology stack and infrastructure

The system also uses publicly available datasets, including wildlife occurrence records from GBIF and iNaturalist, land-cover data from ESA WorldCover, population data from WorldPop, climate information from NASA POWER, and elevation data from Copernicus GLO-30 or other documented sources. Published elephant GPS movement statistics are used to inform the simulated movement component.

The proposed infrastructure separates analytical processing from application-level operations. This arrangement allows the machine-learning and movement-estimation components to generate outputs that can be consumed by the routing component and presented through the common interface.

The final technology versions, package dependencies, hosting environment, and infrastructure configuration must be documented according to the actual prototype implementation.

5.2 Development and integration process

The project follows a component-based development approach in which each component addresses a specific technical problem before being integrated into the overall system. This approach supports independent development and allows the interfaces between components to be defined and evaluated.

5.2.1 Data Collection and Preparation

The initial stage involves identifying and preparing the datasets required by the four components. Wildlife occurrence records support habitat-suitability modelling, while land-cover, population, and climate data contribute to spatial risk estimation. Road-network data supports road-segment analysis and route calculation.

The datasets must be checked for missing values, duplicate records, inconsistent coordinate systems, and unsuitable formats. Spatial datasets must also be transformed into compatible coordinate reference systems where necessary.

5.2.2 Component Development

The four components are developed according to their individual objectives.

- Seasonal Danger Map: Processes public wildlife occurrence and environmental data to estimate spatial HEC risk.

- Sparse Tracker: Uses simulated elephant movement and an Extended Kalman Filter to estimate positions and positional uncertainty.
- Safe Roads: Combines spatial risk, movement estimates, and road-network features to estimate road-segment risk.
- Path Guard: Uses road-network and risk information to calculate and compare shortest-distance and risk-aware routes.

Each component is expected to produce outputs in formats that can be used by the other components. The exact implementation and completion status of each component must be established from the project code and development records.

5.2.3 Interface Development

The integration stage requires consistent data structures and communication interfaces. Spatial risk information is intended to be provided through geospatial files and service interfaces, while movement estimates and road-risk information can be exchanged through structured data representations.

Path Guard uses the road network and road-risk outputs to calculate route alternatives. The common Next.js interface is intended to bring the results together so that users can view risk information and compare available routes.

Interface testing should verify that each component receives the expected inputs, interprets them correctly, and produces outputs in the agreed format.

5.2.4 Integrated Testing and Refinement

After the components are connected, the integrated system should be tested using representative datasets and simulated scenarios. Testing should cover data exchange, spatial consistency, movement-estimation outputs, road-risk calculations, route generation, and presentation of results.

Errors discovered during integration should be documented and corrected where possible. Changes to data formats, model outputs, or interface definitions should be communicated to the components that depend on them.

5.3 Deployment and operational configuration

The proposed deployment model consists of analytical services, a backend application, and a common web interface. These elements are intended to operate together to deliver spatial risk information and route-planning functionality.

5.3.1 Application Configuration

The proposed configuration includes:

1. A Python-based analytical environment for spatial processing, machine-learning models, and movement estimation.
2. A FastAPI service for providing access to spatial risk outputs.
3. A Node.js/Express backend for application-level processing and routing operations.
4. A Next.js frontend for displaying maps, risk information, and route alternatives.
5. Socket.IO for updates and communication where required by the implementation.
6. Local or service-accessible storage for prepared datasets, model outputs, and the road-network representation.

The application should use configuration files or environment variables for service addresses, data paths, and other environment-specific settings. Sensitive configuration values should not be embedded directly in source code.

5.3.2 Data and Model Availability

The system depends on the availability of prepared spatial datasets, model files, and road-network data. These resources should be organised consistently so that the analytical components and routing backend can access the required inputs.

The road-network representation used by Path Guard should remain consistent with the road segments and identifiers used by Safe Roads. Similarly, the spatial risk outputs should use a documented coordinate reference system and data format.

For the prototype, simulated elephant movement should be clearly distinguished from real-world observations. The application should not display simulated estimates as verified live elephant locations.

5.3.3 Operational Limitations

The proposed system is designed as a software-only prototype and does not depend on physical tracking devices or newly installed field sensors. Its operation nevertheless depends on the quality and availability of public data, the successful execution of its analytical services, and the consistency of the integrated components.

The system is not currently established as an operational public warning service. Before real-world deployment, further validation would be required to assess model reliability, data freshness, system performance, user safety, and the suitability of the outputs for decision-making.

The actual hosting platform, installation procedure, deployment status, and operational configuration must be recorded from the implemented prototype rather than assumed.

5.4 Evidence of team execution

The system is a software-only prototype, so its deployment requirements concern software environments, data files, application services and communication between components rather than physical monitoring equipment.

The proposed architecture contains machine-learning functionality, geographic outputs, service APIs, routing logic and a unified interface. These parts must be configured to use compatible data paths, service addresses, dependencies and output formats.

The following table identifies the deployment information that should be documented.

Configuration area	Details to document
Runtime environment	Operating system and relevant language/runtime versions
Python environment	Python version, required libraries and dependency files
Machine-learning models	Model files, input features and model-loading procedure
Geographic data	Dataset locations, coordinate reference systems and preprocessing requirements
API services	Implemented endpoints, ports, configuration and expected responses
Routing service	Road-network file location, graph preparation and route-calculation settings
User interface	Framework version, configuration and method of connecting to backend services
Application startup	The actual steps required to launch each service and the integrated interface

Table 5.3- Deployment and configuration details to record

5.4.1 Configuration and Reproducibility

The project should aim to make the prototype reproducible so that another developer or evaluator can understand how to run the system and repeat its tests.

To support reproducibility, the team should document the required datasets, dependencies, model files and configuration settings. Any data that cannot be

redistributed should be identified, together with the procedure required to obtain it where permitted.

API keys, passwords and other credentials should not be included in the report or committed to a public source repository. If environment variables are used, the report can describe their purpose without disclosing secret values.

5.4.2 Deployment Status

The final report should specify whether the system runs locally, on a shared development machine or through a hosted environment. It should also state which components can be started successfully and whether the integrated interface can communicate with all required services.

The available proposal establishes the intended architecture, but it does not by itself confirm the final hosting environment, deployment procedure or current operational status of every component. These details must be completed from the team's actual implementation.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

The evaluation framework for the Human–Elephant Conflict Early Warning and Safe Routing System focuses on determining whether the four components can work together to provide spatial risk information, estimate elephant movement, assess road-segment risk, and support risk-aware route planning.

The evaluation is organised around the project's four main research questions:

- **RQ1:** Can publicly available wildlife, land-cover, population, and climate data be used to produce a seasonal HEC risk map for Anuradhapura District?

- **RQ2:** How accurately can an Extended Kalman Filter estimate movement along a simulated elephant route when location updates are infrequent?

- **RQ3:** Can road-network information, spatial risk, and movement uncertainty support road-risk assessment and risk-aware route selection?

- **RQ4:** Can the four components be integrated into a common application that presents understandable risk information and route alternatives?

The overall evaluation considers both the performance of individual components and their contribution to the integrated system.

6.1.1 Evaluation Criteria

Evaluation area	Evaluation method	Success evidence
Spatial risk mapping	Examine model evaluation metrics and generated risk maps	Risk outputs are generated and model performance is assessed using documented metrics
Movement estimation	Compare estimated positions with known positions along simulated routes	Position errors and uncertainty coverage are calculated and reported
Road-risk assessment	Examine road-segment outputs and their relationship to the defined proxy labels	Risk values or classifications are produced and interpreted consistently
Route planning	Compare shortest-distance and risk-aware route alternatives	Valid routes are generated and their distances and estimated risks can be compared
Component integration	Test the exchange of data between the four components	Required inputs and outputs are transferred in compatible formats
End-to-end operation	Execute a complete scenario from risk information to route presentation	The integrated workflow produces interpretable outputs for the selected test scenario
Usability and understandability	Conduct structured user evaluation when participants are available	Feedback identifies whether users can understand the displayed risks and route alternatives

Table 6.1-Evaluation framework and success criteria

The evaluation should use documented datasets, test scenarios, metrics, and acceptance criteria. Since completed integrated test records have not been provided, these criteria describe the intended evaluation framework rather than verified test outcomes.

6.2 Integration and end to end testing

Integration testing is intended to verify that the four components exchange data correctly and operate as a connected system. End-to-end testing evaluates the complete workflow from the availability of spatial risk and simulated movement data to the presentation of route alternatives in the common interface.

6.2.1 Interface and Integration Testing

Interface testing should verify the compatibility of component inputs and outputs. Particular attention should be given to the spatial risk data exchanged between the Seasonal Danger Map and Safe Roads, the movement estimates exchanged between the Sparse Tracker and Safe Roads, and the road-network and risk information used by Path Guard.

The following test cases define the main integration checks.

Test ID	Test case	Expected result
INT-01	Load spatial risk outputs	The risk data is read in the expected format and spatial reference system.
INT-02	Transfer movement estimates	The estimated position and uncertainty information are available to Safe Roads.
INT-03	Integrate road-risk outputs	Safe Roads produces road-segment risk information in a format usable by Path Guard.
INT-04	Load the road network	Path Guard can access a compatible road network for route calculation.
INT-05	Generate route alternatives	The system calculates available shortest-distance and risk-aware routes for a valid test scenario.
INT-06	Display integrated results	The common interface presents the available risk information and route alternatives consistently.
INT-07	Handle missing inputs	The system reports missing or invalid data rather than presenting incomplete results as reliable.
INT-08	Verify spatial consistency	Risk cells, movement estimates, and road segments are compared using compatible coordinate systems.

Table 6.2.1- Interface and Integration Testing

These test cases establish the intended integration checks. Their actual results must be recorded after execution.

6.2.2 End-to-End Testing

An end-to-end test should use a representative scenario within the supported study area. The test begins with prepared spatial risk data and a simulated movement route. The resulting movement estimates and uncertainty information are then passed to Safe Roads, which produces road-segment risk information. Path Guard subsequently uses the road network and risk information to generate route alternatives for a selected origin and destination.

The test should verify that:

1. The required input datasets can be accessed.

2. The spatial risk and movement components produce usable outputs.

3. Safe Roads processes the expected inputs and produces road-risk information.

4. Path Guard generates valid route alternatives where a connected route exists.

5. The interface presents the outputs in an understandable form.

6. Missing data or service errors are handled without misleading the user.

6.2.3 Regression, Security, Performance and Usability Testing

Regression testing should be performed after changes to models, data formats, APIs, or routing logic to check that previously working functionality remains available.

Security testing should examine relevant application interfaces, input validation, configuration management, and protection against unauthorised access where applicable. Performance testing should measure processing and response times for representative datasets and route requests. Usability testing should examine whether users can interpret the risk map, understand movement uncertainty, recognise road-risk warnings, and compare route options.

User acceptance testing should evaluate whether the integrated prototype meets its intended functional requirements. No completed regression, security, performance, usability, or user acceptance test results have been supplied for this report. Therefore, these activities cannot be described as successfully completed.

6.3 Overall results

The overall results should demonstrate how the four components contribute to the integrated system. The research proposal provides preliminary component-level evaluation figures for the proposed spatial risk mapping and simulated movement estimation approaches. These figures must be distinguished from independently verified results of the completed integrated prototype.

6.3.1 Spatial Risk Mapping

The research proposal reports a spatial cross-validation AUC of approximately 0.84 and an out-of-fold AUC of approximately 0.85 for the proposed habitat-suitability modelling approach. These values are proposal-reported results and should be included in the final evaluation only after confirming the corresponding experiments, datasets, and calculation procedures.

The risk map is intended to combine habitat suitability, human exposure, and seasonal rainfall information. Its overall usefulness depends not only on the model's discrimination performance but also on the quality of the input records, the suitability of the proxy variables, and the interpretation of the resulting risk scores.

6.3.2 Simulated Movement Estimation

The research proposal reports the following preliminary results for the simulated Sparse Tracker evaluation.

Metric	**Proposal-reported value**
Average position error	Approximately 213 m
Median position error	Approximately 169 m
Error at 15–30 minutes	Approximately 171 m
Error at 45–60 minutes	Approximately 381 m
True position within the 2σ uncertainty circle	Approximately 96.9%

Table 6.3.2- Simulated Movement Estimation

These figures describe the proposed simulation-based evaluation. They do not establish the accuracy of tracking real elephants in Anuradhapura, and they should not be presented as verified integrated-system results without checking the underlying experimental records.

6.3.3 Road-Risk Assessment and Route Planning

The intended outputs of Safe Roads are road-segment risk estimates and warning information. Path Guard is intended to use these outputs to compare shortest-distance and risk-aware routes.

A complete evaluation should report road-risk model metrics, route distances, estimated risk exposure, additional travel distance, and the number of test scenarios in which valid route alternatives were generated. These results are necessary to determine whether risk-aware routing provides a useful trade-off between route length and estimated risk.

No verified integrated road-risk or route-comparison results have been supplied. Consequently, the effectiveness of the complete routing workflow cannot yet be quantified.

6.3.4 Integrated System Results

Evaluation area	Current evidence available for this report	Overall status
Spatial risk modelling	Preliminary metrics reported in the research proposal	Requires verification
Simulated movement estimation	Preliminary simulation metrics reported in the research proposal	Requires verification
Road-risk assessment	Proposed model and proxy-label approach described	Integrated results not supplied
Risk-aware routing	Proposed route-comparison approach described	Integrated results not supplied
End-to-end integration	Proposed component workflow documented	Test evidence not supplied
User validation	No participant feedback or evaluation records supplied	Not established

Table 6.3.4- Integrated System Results

The available information supports the description of the proposed evaluation approach and preliminary component-level results. It is insufficient to establish that the integrated prototype has passed all testing criteria or achieved a measurable reduction in real-world HEC incidents.

6.4 Stakeholder and user validation

Stakeholder and user validation is intended to assess whether the integrated system presents useful and understandable information for people who may need to consider HEC risk when planning journeys or working in affected areas.

Potential evaluation participants include residents of affected rural communities, road users, relevant wildlife or conservation personnel, and other stakeholders with knowledge of local HEC challenges. Participation should be arranged appropriately, and feedback should be collected with informed consent.

A structured evaluation could ask participants to complete representative tasks, such as locating a higher-risk area on the map, interpreting a movement uncertainty indicator, identifying a potentially hazardous road section, and comparing two route alternatives.

Feedback could be collected through task completion records, questionnaires, interviews, or structured observation. The evaluation should examine whether the displayed information is understandable, whether users recognise the uncertainty of the estimates, and whether route comparisons provide sufficient information to support informed decisions.

No completed stakeholder evaluation, participant records, questionnaire responses, or user feedback has been supplied. Therefore, this report cannot claim that the prototype has been validated by community members or relevant authorities.

Before operational use, further consultation would be required to assess the appropriateness of the risk indicators, the clarity of warnings, the suitability of route recommendations, and the consequences of incorrect or outdated information.

6.5 Achievement of common objectives

Common objective	Evidence considered	Assessment
Develop a spatial HEC risk-mapping approach using public data	Proposed methodology and preliminary model metrics described in the research proposal	Partly evidenced; implementation and evaluation require verification
Estimate elephant movement from sparse observations	Proposed EKF method and preliminary simulation metrics described	Partly evidenced; underlying experiments require verification
Assess road-segment risk using spatial and movement information	Proposed model inputs and proxy-label strategy documented	Not yet demonstrated by supplied integrated results
Integrate road-risk information into route planning	Proposed Path Guard architecture and routing method documented	Not yet demonstrated by supplied end-to-end test evidence
Present integrated risk information through a common application	Proposed common interface and system workflow documented	Completion and functionality require verification
Evaluate the integrated system and its usefulness	Evaluation framework and proposed test cases documented	Not yet established by supplied test or user-validation evidence

Table 6.5- Achievement of common objectives

6.6 Integrated limitations and threats

The integrated system has several limitations that affect the interpretation of its outputs and the extent to which the prototype can be applied beyond the research setting.

6.6.1 Data Limitations

Public wildlife occurrence records may be incomplete, spatially biased, or recorded at different levels of accuracy. Land-cover, population, and climate datasets may also differ in spatial resolution and update frequency. These limitations may affect the reliability of the estimated risk map.

6.6.2 Simulation and Movement Uncertainty

The Sparse Tracker uses simulated movement informed by published elephant movement statistics. Movement patterns from African elephant datasets may not fully represent elephant behaviour or landscape conditions in Anuradhapura. The simulation therefore cannot establish the accuracy of the model for real elephants in Sri Lanka.

6.6.3 Proxy-Label Limitations

The Safe Roads component uses proposed proxy labels based on risk corridors and proximity to farmland or settlements. Such labels may not represent actual incident frequency or severity. Model performance against these labels would not, by itself, demonstrate reliable prediction of real human–elephant encounters.

6.6.4 Component Dependencies

The output of each component depends on the quality and availability of upstream information. Errors in the spatial risk map or movement estimates may influence road-risk calculations, which may then affect route recommendations. Incompatible coordinate systems, missing data, inconsistent road identifiers, or unavailable services may also disrupt the integrated workflow.

6.6.5 Evaluation and Generalisability

The proposal-reported component metrics require confirmation against the underlying experiments. Without verified end-to-end testing, it is not possible to establish the complete system's performance, reliability, or practical benefit. Results obtained from the selected study area and simulated scenarios may not generalise to other districts or operating conditions.

6.6.6 Safety, Ethics and Operational Use

The system does not guarantee that a recommended route is safe. Inaccurate, incomplete, or outdated risk information could lead users to make inappropriate decisions if they interpret the output as a confirmed warning.

The prototype should therefore communicate uncertainty clearly and should not be presented as a replacement for official wildlife warnings or emergency guidance. Further field validation, stakeholder consultation, and operational assessment would be required before real-world deployment.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

The proposed Human–Elephant Conflict Early Warning and Safe Routing System aims to provide an accessible, software-based approach to understanding potential HEC risk and comparing route alternatives. Its main value proposition is the integration of spatial risk estimation, simulated elephant movement, road-risk assessment, and risk-aware route planning within a common application.

The system is designed to use publicly available datasets and software-based modelling rather than depending on newly installed physical tracking devices or field sensors. This approach offers a potential starting point for research, demonstration, and future development in areas where access to continuous animal-tracking information may be limited.

The potential beneficiaries include the following groups.

Beneficiary	**Potential value**
Rural residents	Access to spatial information about potential HEC risk areas
Road users	An opportunity to compare route alternatives using estimated road risk
Local communities	A possible tool for improving awareness of geographical and seasonal risk patterns
Wildlife and conservation organisations	A prototype for exploring how public environmental data can support risk assessment
Researchers and students	A framework for studying geospatial modelling, movement estimation, and risk-aware routing
Local authorities and relevant agencies	A potential supplementary decision-support tool, subject to validation and approval

Table 7.1- Value proposition and beneficiaries

The system's principal contribution is the combination of several analytical functions in a connected workflow. However, the benefits listed above are potential benefits rather than demonstrated outcomes. Evidence from stakeholder evaluation and operational testing would be required to establish whether the system provides measurable value to its intended users.

7.2 Market alternatives and differentiation

The proposed system addresses a problem that may be approached through several alternative methods, including official wildlife warnings, community-based reporting, static risk maps, conventional navigation applications, and dedicated wildlife-monitoring technologies.

Static risk maps can help users understand areas associated with potential risk, but they may not account for estimated movement or route-specific risk. Conventional navigation applications generally focus on road connectivity, travel time, and distance rather than the specific HEC risk model proposed in this project. Dedicated animal-tracking systems may provide movement information but can require access to tracking devices, specialised infrastructure, or restricted data.

The proposed system seeks to differentiate itself through the integration of four functions:

- Spatial HEC risk estimation using public environmental and wildlife datasets.
- Movement estimation using sparse observations and simulated movement.
- Road-segment risk assessment using spatial risk and movement uncertainty.
- Route comparison that considers estimated risk alongside travel distance.

The software-only approach also offers a potential advantage for research and prototyping because it does not require the project itself to install physical tracking equipment.

Nevertheless, the prototype has important limitations compared with established operational services. It does not have demonstrated real-time tracking of local elephants, verified incident-based road-risk predictions, or proven reductions in HEC incidents. Its differentiation should therefore be understood as an architectural and research contribution rather than an established commercial advantage.

7.3 Business model and feasibility

7.3.1 Potential Delivery Models

Model	Description	Main consideration
Research and educational tool	Used for demonstrations, academic projects, and further modelling research	Requires documented methods and reproducible evaluation
Community information platform	Provides accessible risk maps and route comparisons for users in affected areas	Requires clear communication, usability testing, and reliable data
Institutional decision-support service	Supports analysis by relevant wildlife, conservation, or local authorities	Requires validation, institutional collaboration, and appropriate authorisation
Open-source software project	Makes suitable parts of the implementation available for review and further development	Requires clear licensing, documentation, and maintenance arrangements

Table 7.3.1- Potential Delivery Models

7.3.2 Resources and Costs

Potential resources required for further development include software development time, data preparation, computing resources, testing infrastructure, interface design, stakeholder consultation, and ongoing maintenance.

Using publicly available datasets and open-source software may reduce certain licensing and infrastructure costs. However, costs may still arise from hosting, data processing, software maintenance, field validation, security, user support, and long-term service operation.

A reliable financial assessment cannot be produced without a defined deployment model, cost estimates, resource requirements, and evidence of demand.

7.3.3 Feasibility Assessment

The proposed architecture is based on established software development approaches, including geospatial processing, machine learning, movement estimation, APIs, and graph-based routing. These technologies provide a plausible technical foundation for a prototype.

However, technical feasibility at the architectural level does not establish operational readiness. Further work is needed to verify the integrated implementation, evaluate performance using representative data, establish the reliability of risk estimates, and assess the needs of potential users.

No confirmed customers, funding arrangements, commercial partners, revenue model, or cost analysis have been provided. Commercial viability therefore remains undetermined.

7.4 Adoption intellectual property and compliance

The adoption of the system would depend on its technical reliability, usability, credibility, accessibility, and suitability for the needs of its intended users.

7.4.1 Adoption Considerations

Potential barriers include limited internet access in some rural areas, differences in users' digital literacy, uncertainty about the meaning of modelled risk scores, and concerns about relying on an unvalidated route recommendation.

The interface should present risk information in understandable language and distinguish estimated risk from confirmed elephant presence. It should also explain that the absence of a warning does not guarantee the absence of danger.

Consultation with affected communities and relevant authorities would help determine whether the interface, warning categories, and route recommendations are appropriate for practical use.

7.4.2 Intellectual Property and Licensing

The project uses or proposes to use datasets, software frameworks, and analytical libraries from external sources. Before public distribution or commercial use, the team should review the relevant licences and attribution requirements for datasets, software dependencies, map data, and any reused movement data.

The ownership and licensing of the team's original source code, models, documentation, and interface should be determined according to the institution's applicable policies and any agreements among team members.

No specific ownership agreement or final software licence has been supplied for this report.

7.4.3 Data Protection and Responsible Data Use

The system is designed around public geographical and environmental datasets and simulated elephant movement. Nevertheless, future versions may process user-selected locations, route requests, or community-submitted reports.

If such information is collected, the project should minimise unnecessary data collection, protect stored information, restrict access where appropriate, and explain how information is used. Community-submitted reports should be handled carefully to reduce the risk of exposing sensitive locations or personal information.

7.4.4 Safety and Professional Responsibility

Incorrect risk estimates or misleading route recommendations could create risks for users. The system should therefore be presented as a supplementary research prototype rather than a guaranteed safety service.

Before deployment, the project would require suitable validation, transparent documentation of limitations, appropriate security measures, and consultation with relevant stakeholders. Any operational deployment should also comply with applicable legal requirements, institutional policies, and the terms of the datasets and software used.

7.5 Sustainable Development Goal contribution

The proposed system has a potential relationship with the United Nations Sustainable Development Goals (SDGs), particularly SDG 15, SDG 11, and SDG 9.

7.5.1 SDG 15 — Life on Land

SDG 15 focuses on protecting terrestrial ecosystems, managing forests sustainably, combating land degradation, and halting biodiversity loss. The project's spatial risk mapping and movement-estimation approach may support research into the relationship between elephant movement, land use, and human activity.

The prototype could contribute to improved understanding of potential conflict areas and support future conservation-related analysis. However, it does not independently demonstrate a reduction in harm to elephants or an improvement in biodiversity outcomes.

7.5.2 SDG 11 — Sustainable Cities and Communities

SDG 11 focuses on making settlements inclusive, safe, resilient, and sustainable. The project's emphasis on rural communities and route-risk information is relevant to the broader goal of improving awareness of environmental risks affecting people and settlements.

The system may provide a foundation for exploring risk-aware route planning in areas affected by HEC. Its actual contribution to community safety would require usability evaluation, local validation, and evidence that the information supports appropriate decisions.

7.5.3 SDG 9 — Industry, Innovation and Infrastructure

SDG 9 concerns resilient infrastructure, sustainable industrialisation, and innovation. The project applies software engineering, geospatial analysis, machine learning, and route-planning methods to a local environmental challenge.

Its modular architecture provides a basis for further technical research and innovation. Any wider impact would depend on the successful implementation, maintenance, validation, and responsible adoption of the system.

Overall, the project offers a potential software-based contribution to conservation research, community risk awareness, and technological innovation. These contributions should be evaluated through evidence rather than assumed from the proposed architecture alone.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

Human–Elephant Conflict remains a significant challenge in rural Sri Lanka, where elephant movement and human activities can overlap and create risks for both communities and wildlife. Limited access to timely, location-specific risk information creates a need for approaches that can combine available environmental data, geographical information, and movement estimates to support informed decisions.

This project proposes an integrated, software-only Human–Elephant Conflict Early Warning and Safe Routing System for Anuradhapura District. The proposed framework combines four components: Seasonal Danger Map, Sparse Tracker, Safe Roads, and Path Guard.

The Seasonal Danger Map estimates potential HEC risk using wildlife occurrence, land-cover, population, and seasonal climate information. The Sparse Tracker estimates elephant movement and positional uncertainty using simulated routes and an Extended Kalman Filter. Safe Roads uses spatial risk, movement estimates, and road-network characteristics to assess potential road-segment risk. Path Guard uses the resulting information to compare shortest-distance and risk-aware routes through a common application.

The principal strength of the proposed solution is its integrated structure. Rather than treating spatial risk mapping, movement estimation, road-risk analysis, and route planning as separate tasks, the architecture connects them into a common workflow. Its software-only design also provides a basis for research and prototyping without requiring the installation of physical tracking devices.

However, the proposed architecture and preliminary component-level results do not, by themselves, establish the effectiveness of the complete system. The available documentation does not include sufficient verified integration tests, end-to-end results, or stakeholder feedback to conclude that the prototype has achieved all its objectives. Its performance and practical usefulness must therefore be established through further implementation verification and evaluation.

The project provides a foundation for investigating how public data, simulated movement, machine learning, and risk-aware routing can be combined to support HEC risk awareness. It should be regarded as a research prototype rather than a validated operational warning service.

8.2 Principal integrated contribution

The principal contribution of the project is the proposed integration of four complementary software components into one framework for HEC risk awareness and route planning.

First, the system connects spatial risk estimation with movement information. This allows the proposed architecture to consider both the geographical distribution of potential risk and the uncertainty associated with estimated elephant movement.

Second, the framework extends spatial risk information to road-segment assessment. By combining geographical risk, movement estimates, and road-network data, the Safe Roads component is designed to provide information that can be used directly in route planning.

Third, Path Guard connects these analytical outputs to a user-oriented task. Instead of presenting only a risk map, the proposed system compares route alternatives and considers estimated risk alongside route distance.

Finally, the project demonstrates a modular architectural approach that can be developed and evaluated using publicly available datasets and simulated movement. This creates a basis for future research into the technical integration of geospatial modelling, movement estimation, and risk-aware navigation.

The value of this contribution lies in the connected design and its potential for further development. Claims of improved safety, reduced HEC incidents, or operational accuracy require additional evidence and should not be inferred from the architecture alone.

8.3 Future directions

Future work should prioritise improvements that strengthen the reliability, evaluation, and practical suitability of the integrated system.

8.3.1 Complete and Verify the Integrated Implementation

The first priority is to verify that all four components operate together as intended. This includes checking API communication, data formats, coordinate reference systems, road-network consistency, error handling, and presentation of results through the common interface.

Integration records and test evidence should be maintained so that the completed implementation can be demonstrated and reproduced.

8.3.2 Strengthen Model Evaluation

The Seasonal Danger Map should be evaluated using documented validation procedures and suitable spatial evaluation methods. Further investigation should examine the influence of data quality, spatial bias, and seasonal variables on risk estimates.

The Sparse Tracker should be tested against known simulated trajectories using documented movement scenarios and update intervals. Its uncertainty estimates should also be evaluated to determine whether the reported uncertainty appropriately reflects position errors.

The Safe Roads component requires further evaluation of its proxy labels and risk predictions. Where reliable incident data becomes available through appropriate access arrangements, it could support stronger validation of the relationship between estimated road risk and observed HEC incidents.

8.3.3 Evaluate Risk-Aware Route Planning

Path Guard should be evaluated across multiple origin–destination pairs and representative simulated risk scenarios. The evaluation should compare shortest-distance routes with risk-aware alternatives using route length, estimated risk exposure, additional travel distance, and route-generation success.

These comparisons would help establish whether the routing method provides a meaningful trade-off between travel distance and estimated risk.

8.3.4 Improve Data and Movement Representation

Future development could investigate the use of more locally representative elephant movement data where such data can be obtained lawfully and appropriately. Updated environmental datasets, improved wildlife occurrence records, and validated information about HEC incidents could also improve the relevance of spatial risk estimates.

Any use of sensitive wildlife location information should follow appropriate data-access, conservation, and ethical requirements.

8.3.5 Conduct Stakeholder and Usability Evaluation

The common interface should be evaluated with potential users and relevant stakeholders. Evaluation should examine whether users understand the risk map, movement uncertainty, road warnings, and route comparisons.

Feedback could guide improvements to map legends, warning messages, risk explanations, and the presentation of uncertainty. Consultation with relevant wildlife and local authorities would also help determine the conditions under which the system might be suitable for supplementary decision support.

8.3.6 Assess Operational Readiness and Sustainability

Before deployment beyond the research prototype, the system requires further assessment of reliability, security, accessibility, data maintenance, hosting costs, and operational responsibilities.

Future work should also determine how the application would obtain updated information, handle service failures, communicate uncertainty, and avoid presenting incomplete information as a reliable warning. Any real-world deployment should be undertaken with appropriate validation and consultation with relevant stakeholders.

In conclusion, the proposed integrated system establishes a structured approach to combining spatial HEC risk assessment, simulated movement estimation, road-risk analysis, and risk-aware route planning. Further implementation verification, systematic testing, local validation, and stakeholder evaluation are necessary to determine its effectiveness and suitability for practical use in Sri Lanka.

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APPENDICES

Appendix A Integrated interface specification

A.1 Purpose

This appendix defines the data exchanged between the four components of the GAJA SAVIYA Human–Elephant Conflict (HEC) Early Warning and Safe-Routing System. The interface specification establishes the expected input data, output data, spatial conventions, validation requirements, and error-handling behaviour needed to support integration.

The integrated system consists of four components: Seasonal Danger Map, Sparse Tracker, Safe Roads, and Path Guard. The Seasonal Danger Map produces spatial risk information, while Sparse Tracker estimates elephant movement and positional uncertainty from simulated movement data. Safe Roads combines these outputs with road-network information to estimate road-segment risk. Path Guard uses the road network and risk information to calculate and compare route alternatives.

The interfaces described below represent the common data contract. The final endpoint names and payload fields must correspond to the implemented application and source code.

A.2 Component interfaces

Interface	Input	Expected output
INT-01: Seasonal Danger Map	Wildlife occurrence data, land-cover data, population data, climate data, and configured spatial parameters	Geographic risk layer, risk attributes, and explanatory information
INT-02: Sparse Tracker	Simulated movement observations, timestamps, filter configuration, and initial state	Estimated position, estimated trajectory, timestamp, and positional uncertainty

INT-03: Safe Roads	Geographic risk layer, movement estimates, uncertainty information, and road network	Road-segment identifiers, risk estimates, and warning classifications
INT-04: Path Guard	Road graph, origin, destination, and available road-risk information	Shortest-distance route, risk-aware route where available, route distance, and associated risk information
INT-05: Common user interface	Available outputs from the four components	Map layers, movement estimates, road-risk information, route alternatives, and error messages

Table A.1- Integrated component interfaces

A.3 Common data conventions

The following conventions should be applied consistently across component interfaces.

1. Spatial reference systems: The project uses EPSG:5234 for its 500 m spatial grid. Coordinate reference systems must be recorded with spatial datasets and transformed when required. GeoJSON exchanged through standard-compliant interfaces uses WGS 84 longitude and latitude coordinates, as specified by RFC 7946 [10].

2. Timestamps: Timestamps should use ISO 8601 date-time notation and include a time-zone designation. Movement estimates and observations must retain their associated timestamps.

3. Identifiers: Every road segment must have a stable identifier that is preserved between Safe Roads and Path Guard. Grid cells and movement records should also have identifiable keys where required.

4. Risk values: Risk values must be accompanied by documented meanings and scales. If the implementation normalises risk to the interval [0,1], this must be stated explicitly. The system must not assume that values from different components share the same scale unless a documented transformation is applied.

5. Missing data: Missing or invalid inputs must be reported. The system must not silently interpret missing risk or movement information as zero risk.

6. Simulation status: Simulated elephant positions must be distinguishable from verified observations. The interface must not present simulated locations as confirmed real-time locations of wild elephants.

A.4 Data structures

The following tables define the minimum logical fields required for integration. They should be aligned with the actual schemas used by the application.

A.4.1 Seasonal Danger Map output

Field	**Type**	**Description**
cell_id	String or integer	Unique identifier of the spatial grid cell
geometry	Polygon or raster-cell reference	Spatial extent of the cell
habitat_suitability	Numeric	Modelled habitat-suitability value
human_exposure	Numeric	Estimated human-exposure value
seasonal_multiplier	Numeric	Seasonal climate adjustment used by the model
risk_score	Numeric	Calculated geographic risk score

season	String	Season or period represented by the output
crs	String	Coordinate reference system of the spatial output

Table A.4.1- Seasonal Danger Map output

A.4.2 Safe Roads output

Field	**Type**	**Description**
segment_id	String or integer	Stable road-segment identifier
geometry	LineString or equivalent reference	Road-segment geometry
segment_risk	Numeric	Estimated road-segment risk
risk_class	String	Display classification, where implemented
distance_to_estimate	Numeric, where available	Distance from the estimated position or uncertainty area
timestamp	Date-time string, where applicable	Time associated with movement-related risk information
risk_basis	String	Indicates whether the classification is modelled, proxy-based, or derived from distance rules

Table A.4.2- Safe Roads output

Road-risk outputs must preserve the segment identifiers required by Path Guard. Proxy-based risk classifications must not be described as confirmed historical HEC incidents.

A.4.3 Path Guard route request and response

A route request must identify the origin and destination and specify the supported routing options. A route response should contain the route geometry, route distance, and the available risk-related information.

Field	Type	Description
origin	Coordinate or supported location identifier	Starting location
destination	Coordinate or supported location identifier	Destination
routing_mode	String	Shortest-distance or risk-aware routing
route_id	String	Identifier for the calculated route
geometry	LineString or equivalent reference	Route geometry
distance	Numeric	Route distance with documented units
estimated_risk	Numeric or documented summary	Risk estimate, where calculated
status	String	Success, no route, invalid input, or other documented result

Table A.4.3- Path Guard route request and response

When the implementation uses a risk-weighted route cost, the cost function and parameter values must be documented. The report describes the following proposed form:

$$C = L(1 + \lambda R)$$

where C is the route cost, L is road-segment length, R is estimated segment risk, and λ controls the influence of risk on routing. The implementation must document how these quantities are scaled and how λ is selected.

A.4 Validation and error handling

Each interface must validate its inputs before processing. At a minimum, the integrated system should check:

- Required fields are present and have the expected types.
- Coordinates are valid and are interpreted using the documented coordinate reference system.
- Timestamps use the agreed format.
- Risk values conform to the documented scale.
- Road-segment identifiers remain consistent between Safe Roads and Path Guard.
- Required data files and services are available.
- Route requests contain valid origin and destination values.
- Missing, malformed, or unavailable inputs produce clear error messages.

The system should distinguish between an invalid request, unavailable data, a failed service, and a valid request for which no connected route exists. Failure in one component must not cause the interface to display incomplete results as reliable warnings.

A.5 Integration verification

The interfaces should be verified through documented integration tests. The tests should establish that each component can read the expected input, produce its defined output, and pass that output to the next component without losing identifiers, timestamps, coordinate information, or risk attributes.

The end-to-end test should use a documented simulation scenario. It should check the flow from spatial risk mapping and movement estimation through road-risk assessment and route calculation to the common user interface.

The actual test outcomes, including failures and corrective actions, should be reported in Chapter 6. This specification defines the intended integration requirements; it does not, by itself, establish that every interface has passed testing.

Appendix B Team contribution declaration

B.1 Purpose

This appendix records the allocation of responsibilities and the declaration of contributions to the common integrated solution titled “GAJA SAVIYA – An Open-Data, Software-Only Early Warning and Safe-Routing Framework for Human–Elephant Conflict in Rural Sri Lanka.”

The common report covers the shared research problem, the four-component architecture, integration, implementation, evaluation, and overall project outcomes. Each member is responsible for accurately reporting their own contribution and confirming the accuracy of the team’s shared record.

B.2 Team members and component responsibilities

The names and student IDs below follow the team information shown on the report’s title page. The component allocation must be confirmed by the team before signing, because the current title page appears to contain an inconsistent component label.

Team member	Student ID	Primary component responsibility
Saisuhani L	IT22091598	Component 3 — Safe Roads

Wijesooriya W.M.R.A.B	IT22249098	Component 2 — Sparse Tracker
Dhananjaya A.G.G	IT22145734	Component 1 — Seasonal Danger Map*
Gunathilake D.R.G.L	IT22544636	Component 4 — Path Guard

Table B.1- Team members and primary component responsibilities

Verification required: the existing title page labels Dhananjaya A.G.G's component as "Component 1 — Safe Roads," while the report's component mapping identifies Safe Roads as Component 3 and Seasonal Danger Map as Component 1. The entry above follows the component numbering used in the body of the report and must be confirmed by the team before submission.

B.3 Individual contribution records

Each member should confirm the tasks they actually completed. The descriptions below identify the expected areas of responsibility; they must be amended if the actual work allocation differed.

Team member	Contribution areas to confirm
Saisuhani L	Development and evaluation of Safe Roads; preparation of road-segment risk outputs; integration of road-risk information with Path Guard
Wijesooriya W.M.R.A.B	Development and evaluation of Sparse Tracker; implementation or evaluation of simulated movement estimation and uncertainty outputs; provision of movement information for integration
Dhananjaya A.G.G	Development and evaluation of Seasonal Danger Map; preparation of geographic risk outputs and supporting data; provision of risk information for integration

Gunathilake D.R.G.L	Development and evaluation of Path Guard; route calculation and comparison; integration of road-network and risk information into the user interface

Table B.3- Individual contribution records

These records are proposed from the component structure described in the report. Each member must verify the tasks attributed to them and correct any inaccuracies before signing.

B.4 Shared team contributions

The following activities concern the common integrated solution and should be confirmed against the team's actual records:

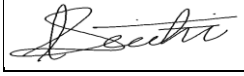
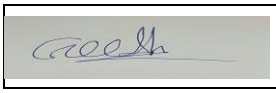
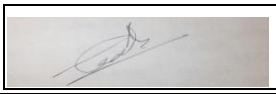
1. Defining the shared research problem, project scope, and objectives.
2. Designing and reviewing the overall four-component architecture.
3. Agreeing on data formats, spatial conventions, identifiers, and component interfaces.
4. Integrating the available component outputs into the common application.
5. Preparing and reviewing the common report and its supporting figures and tables.
6. Conducting integration and end-to-end testing, where completed, and recording the actual results.
7. Reviewing limitations, responsible use, and the distinction between simulated outputs and verified real-world observations.

Activities that were not completed must not be declared as completed. Supporting records, such as source-control history, meeting notes, test reports, or task-board records, should be retained where available.

B.5 Declaration of accuracy and shared responsibility

We confirm that the contribution information recorded in this appendix accurately represents our individual and shared work after review by all team members. We understand that the common report describes the integrated solution and that all team members share responsibility for the accuracy of the common claims, citations, data descriptions, reported results, and conclusions.

We further confirm that individual contributions have been represented fairly and that no uncompleted task, test, result, or activity has been declared as completed. Any changes to the allocation of responsibilities have been reviewed and agreed upon by the relevant members.

Team member	Signature	Date
Saisuhani L		11/10/2026
Wijesooriya W.M.R.A.B		11/10/2026
Dhananjaya A.G.G		11/10/2026
Gunathilake D.R.G.L		11/10/2026